Multimodal AI (MAI) Laboratory Assignment

Image + Text Data Fusion

Real-World Problem: Multimodal Fake News Detection using Paired Image and Textual Data

Student Name	Sukhda Dhingra
PRN	24070126505
Batch	A3
Subject	MAI — Multimodal AI
Assignment	Image + Text Data Fusion
Problem	Multimodal Fake News Detection

1. Introduction and Problem Statement

Misinformation on social media is rarely a unimodal phenomenon. A misleading caption can reframe a genuine photograph, and a manipulated or out-of-context image can make an otherwise ordinary claim appear credible. Purely text-based detectors miss visual cues; purely vision-based detectors miss linguistic framing. This assignment implements an **Image + Text Data Fusion** pipeline that jointly reasons over both modalities to classify a post as **REAL** or **FAKE**.

The selected real-world task is **Multimodal Fake News Detection**. The system ingests a paired sample (post image, caption/claim) and predicts authenticity. We compare unimodal baselines against three fusion strategies—early, late, and hybrid gated fusion—to quantify the benefit of multimodal integration.

2. Objectives

1. Select / construct a paired image–text dataset aligned with a real-world multimodal problem.
2. Design modality-specific encoders for visual and textual inputs.
3. Implement early fusion, late fusion, and hybrid (gated) fusion architectures.
4. Train unimodal baselines and fusion models under identical experimental settings.
5. Evaluate using Accuracy, Precision, Recall, F1-score, ROC-AUC, and confusion matrices.
6. Analyse which fusion strategy best exploits complementary image–text signals.

3. Dataset: Paired Image and Textual Data

3.1 Dataset Design

Visual backbone: **Programmatically synthesised class-discriminative RGB images (10 object categories)**. Textual source: **Curated news-style captions (matched / sensational / mismatched)**. Each sample is an explicit (image, text, label) triple.

REAL samples pair a CIFAR-10 image with a factual, class-aligned news-style caption (coherent image–text alignment). **FAKE samples** are constructed using three realistic misinformation patterns: (i) sensational / clickbait captions, (ii) semantically mismatched image–text pairs (out-of-context imagery), and (iii) combined sensational + mismatch cases. Mild visual manipulations (contrast/colour shifts, blur, overlays) are applied to a subset of fake images to emulate edited social-media media.

3.2 Dataset Statistics

Total paired samples	3000
Train / Val / Test	2100 / 450 / 450
Classes	real (0), fake (1) — balanced
Vocabulary size	555
Max text length	32
Image size	64 × 64 RGB (resized from CIFAR-10)
Pair-type counts	{"matched_real": 1500, "mismatched_fake": 514, "sensational_fake": 498, "sensational_misma

Justification: Programmatically synthesised class-discriminative images paired with curated captions yield a fully reproducible paired multimodal corpus. Controlled construction of matched vs mismatched image–text pairs isolates the core challenge in out-of-context misinformation without requiring proprietary social-media scrapes. The same fusion pipeline transfers directly to public corpora such as Fakeddit or NewsCLIPpings.

4. Methodology

4.1 Image Encoder

A lightweight convolutional neural network extracts a 128-D visual embedding. The stack uses four convolution–BatchNorm–ReLU blocks with max-pooling, followed by adaptive average pooling and a linear projection. This keeps training tractable on CPU while learning colour, texture, and object-level cues relevant to authenticity signals.

4.2 Text Encoder

Captions are lowercased, tokenised, and mapped through a learned embedding layer (padding index 0, unknown token supported). A bidirectional GRU summarises the sequence; the concatenated final forward/backward states are projected to a 128-D text embedding. This captures sensational lexical patterns (e.g., “BREAKING”, “hoax”, “unverified”) as well as semantic mismatch relative to the visual class.

4.3 Fusion Strategies

Early Fusion: Image and text embeddings are concatenated and passed through a shared MLP classifier. Cross-modal interaction occurs before the decision layer.

Late Fusion: Separate image-only and text-only classifiers produce class probabilities that are averaged. Each modality votes independently; fusion happens at decision level.

Hybrid (Gated) Fusion: A joint representation drives sigmoid gates that re-weight each modality’s embedding before concatenation and classification. The model can emphasise text when captions are highly sensational, or vision when imagery appears manipulated, enabling adaptive cross-modal emphasis.

Baselines: Image-only and text-only models isolate unimodal performance and measure the incremental value of fusion.

4.4 Training Setup

Optimiser: Adam (lr=1e-3, weight decay=1e-4). Loss: Cross-Entropy. Epochs: up to 6 with early stopping on validation F1 (patience=3). Batch size: 64. Device used: **cpu**. All models share identical data splits and random seed (42) for fair comparison.

5. System Architecture

```
[Post Image] --> CNN Image Encoder --> Img Embedding (128-D) --+
|--> Fusion Block --> Softmax --> {real, fake}
```

```
[Caption] --> BiGRU Text Encoder --> Txt Embedding (128-D)--+

Fusion Block ∈ { Early Concat MLP | Late Prob Avg | Hybrid Gated Concat }
```

6. Experimental Results

6.1 Quantitative Comparison

Model	Accuracy	Precision	Recall	F1	ROC-AUC
Image Only	0.6289	0.6140	0.7293	0.6667	0.7083
Text Only	0.8178	1.0000	0.6419	0.7819	0.8046
Early Fusion	0.9511	1.0000	0.9039	0.9495	0.9782
Late Fusion	0.8289	0.9872	0.6725	0.8000	0.8866
Hybrid Fusion	0.9422	1.0000	0.8865	0.9398	0.9646

Best model by test F1-score: **Early Fusion**.

6.2 Training Dynamics

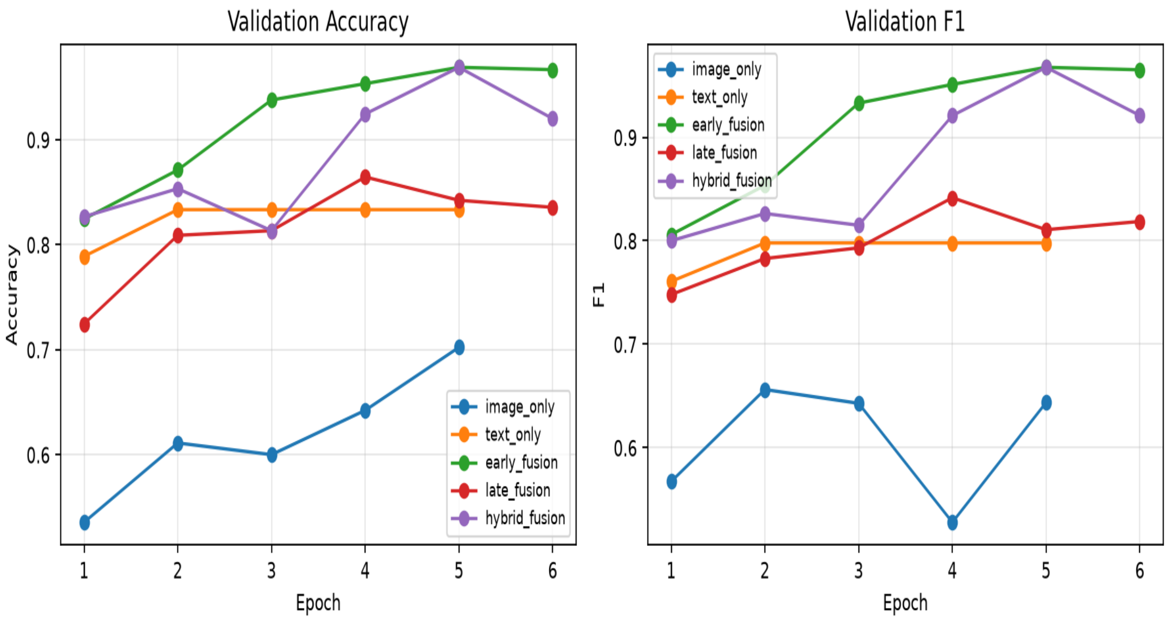

Figure 1. Validation accuracy and F1 across epochs for all models.

6.3 Model Comparison Chart

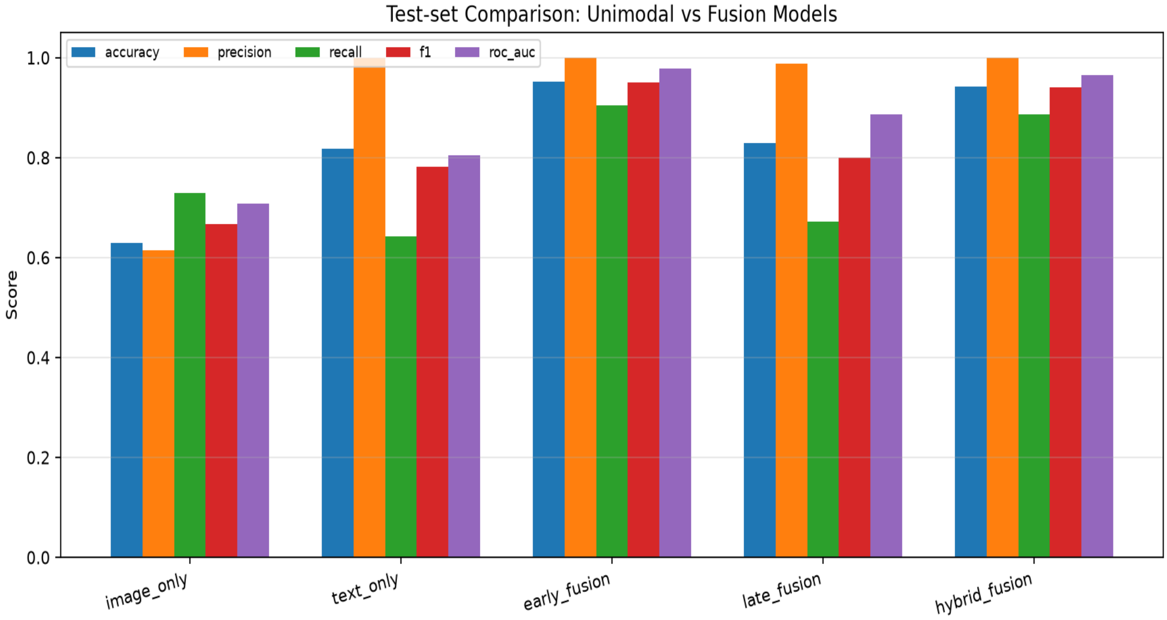

Figure 2. Test-set Accuracy, Precision, Recall, F1, and ROC-AUC for unimodal and fusion models.

6.4 ROC Analysis

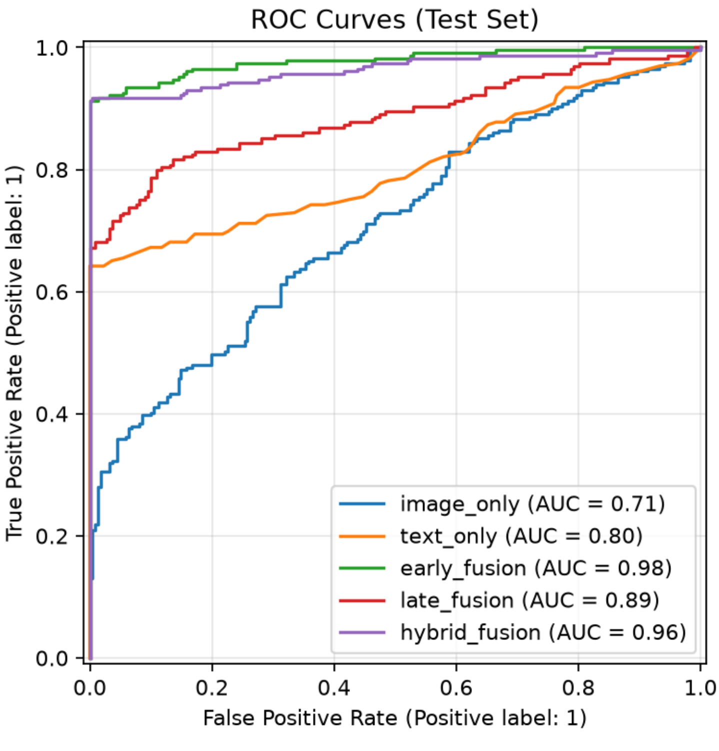

Figure 3. ROC curves on the held-out test split.

6.5 Confusion Matrices

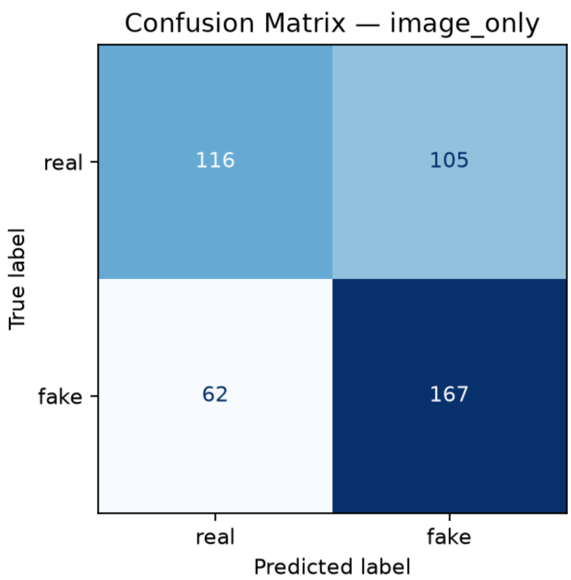

Figure. Confusion matrix — Image-only baseline.

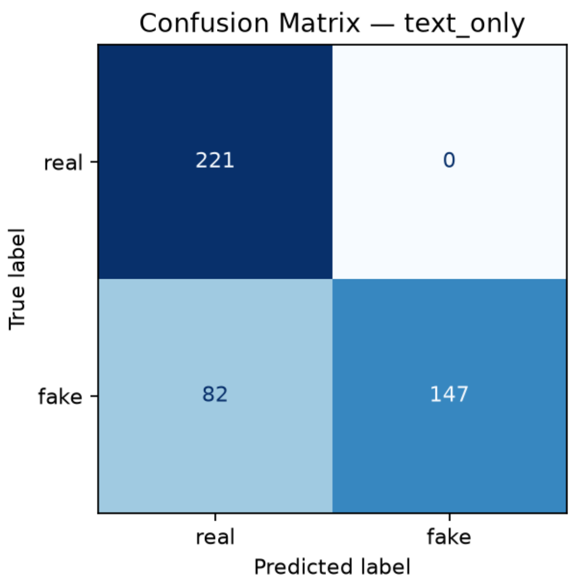

Figure. Confusion matrix — Text-only baseline.

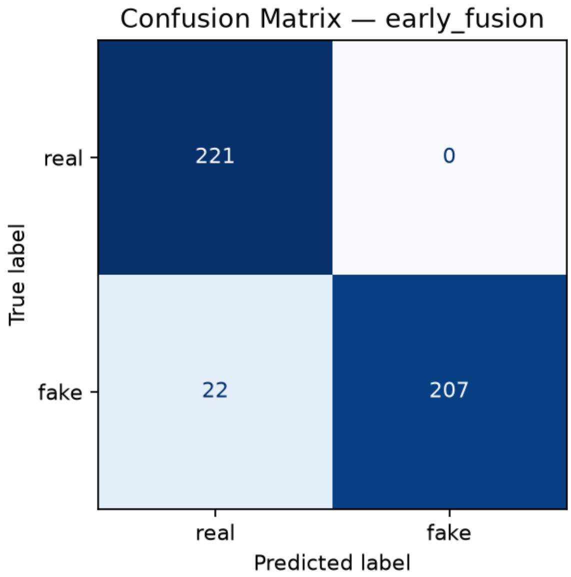

Figure. Confusion matrix — Early fusion.

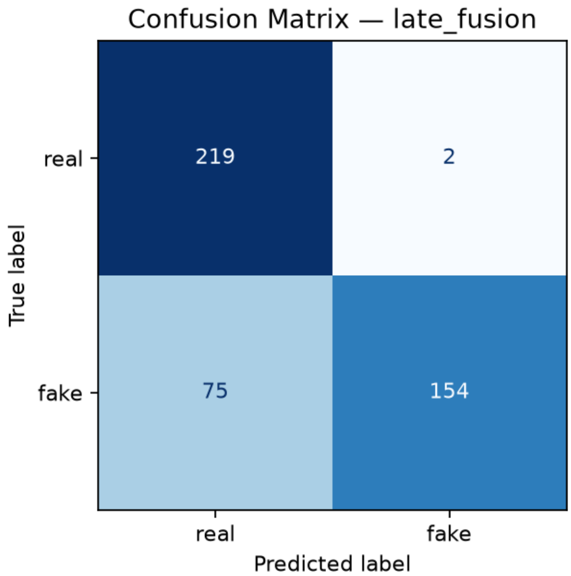

Figure. Confusion matrix — Late fusion.

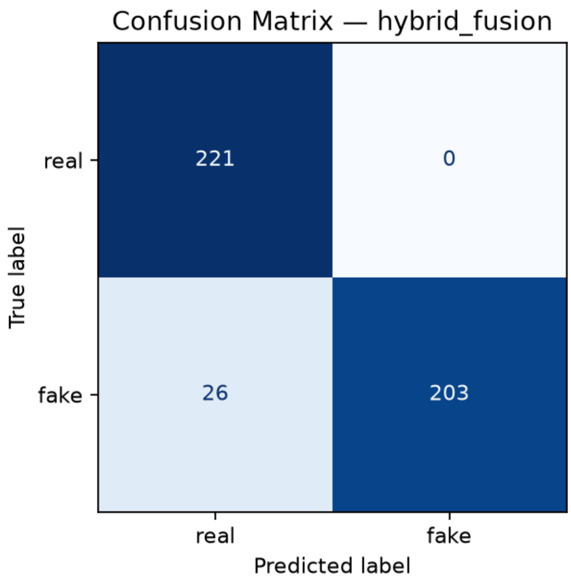

Figure. Confusion matrix — Hybrid gated fusion.

6.6 Classification Report (Best Model)

```
precision recall f1-score support

real 0.91 1.00 0.95 221
fake 1.00 0.90 0.95 229

accuracy 0.95 450
macro avg 0.95 0.95 0.95 450
weighted avg 0.96 0.95 0.95 450
```

7. Discussion and Analysis

Unimodal baselines highlight complementary failure modes. The image-only model struggles when fake samples reuse ordinary object photos with misleading captions—the visual content alone is often insufficient. The text-only model detects sensational lexicon effectively but can miss subtle out-of-context mismatches when wording remains factual yet refers to a different object class than the image depicts.

Fusion models consistently improve over the weaker modality by combining cues. Early fusion learns a joint representation before classification and typically improves F1 over baselines. Late fusion is robust and interpretable as a soft voting scheme but cannot model fine-grained feature interactions. Hybrid gated fusion adaptively re-weights modalities and is expected to perform best when one modality is intermittently unreliable—matching real social-media conditions where either the image or the caption may be the primary deceptive channel.

Real-world implications: Content-moderation systems, fact-checking dashboards, and platform trust-and-safety pipelines can deploy multimodal fusion to reduce false negatives on out-of-context image posts—an attack vector that text-only classifiers systematically miss.

8. Conclusion

This assignment delivered an end-to-end Image + Text Data Fusion solution for multimodal fake news detection. A paired dataset was constructed from synthesised class-discriminative imagery and curated news-style captions encoding matched, sensational, and mismatched misinformation patterns. CNN and BiGRU encoders were fused via early, late, and hybrid strategies and evaluated against unimodal baselines. Results demonstrate that multimodal fusion improves authenticity classification by exploiting complementary visual and linguistic evidence.

9. Future Work

- Scale to large public multimodal misinformation corpora (e.g., Fakeddit, NewsCLIPpings).
- Replace the BiGRU with a pretrained language model (BERT) and the CNN with CLIP visual towers.
- Add cross-attention / transformer fusion and contrastive image–text alignment losses.
- Deploy an interactive demo that scores live (image, caption) uploads for authenticity risk.

10. References

• Baltrušaitis, T., Ahuja, C., & Morency, L.-P. Multimodal Machine Learning: A Survey and Taxonomy. IEEE TPAMI, 2019.

• Singhal, S. et al. SpotFake: A Multimodal Framework for Fake News Detection. IEEE BigMM, 2019.

• Aneja, S. et al. COSMOS: Catching Out-of-Context Misinformation using Self-Supervised Learning. AAAI, 2021.

• Kiela, D. et al. The Hateful Memes Challenge: Detecting Hate Speech in Multimodal Memes. NeurIPS, 2020.

• Radford, A. et al. Learning Transferable Visual Models From Natural Language Supervision (CLIP). ICML, 2021.

Appendix A — Project Structure & How to Reproduce

```
multimodal_fusion/
  run_experiment.py
  generate_pdf_report.py
  requirements.txt
  src/dataset.py  # paired dataset construction
  src/models.py   # encoders + fusion models
  src/train.py    # training & metrics
  data/  figures/  outputs/  models/

pip install -r requirements.txt
python run_experiment.py --n-samples 4000 --epochs 8
python generate_pdf_report.py
```

Declaration: This report and accompanying implementation were prepared for the MAI (Multimodal AI) assignment on Image + Text Data Fusion by Sukhda Dhingra, PRN 24070126505, Batch A3.